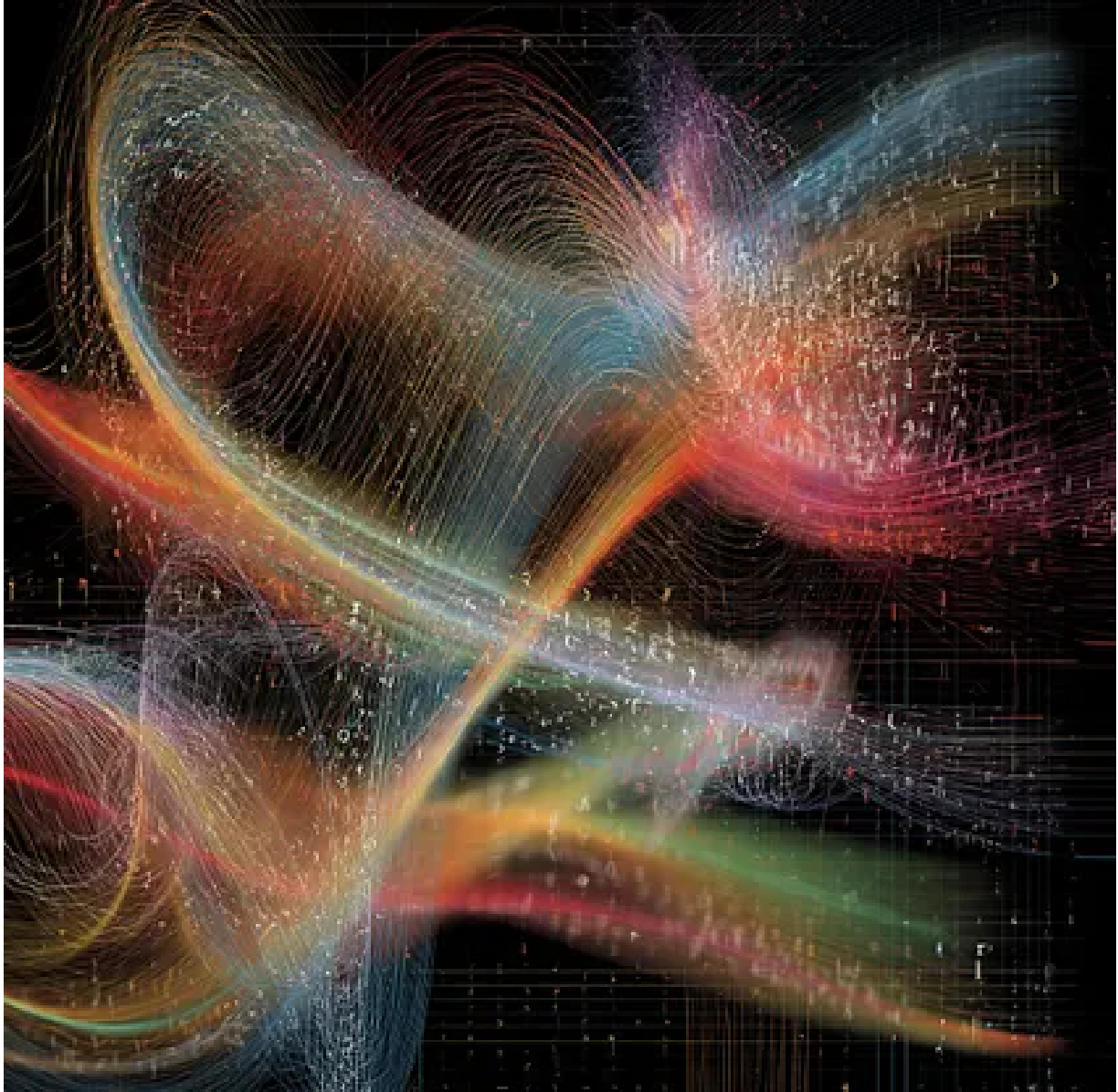
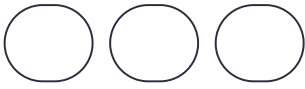


Introducing Functional Symbolic Trajectories

Completing the Picture of Language as a Nonlinear Dynamical System

KEVIN R. HAYLETT

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The Dynamics of Language

In my earlier essay, “[Geofinitism: Language as a Nonlinear Dynamical System](#) (January 2026)” I described language not as a collection of fixed meanings, but as a dynamic process unfolding in real time.

Words and sentences form a dynamical time series. In speech, sound waves arrive one after another. In writing, symbols are read one after another, usually from left to right and line by line. Meaning does not appear all at once. It is built through sequence, memory, expectation, correction, and reconstruction.

This is not simply a metaphor. It is a modelling claim grounded in the mathematics of nonlinear dynamical systems. That is the mathematical study of how complex systems change over time, how they settle into patterns, how they drift, how they become unstable, and how small changes can sometimes redirect the whole path.

The earlier work argued that reading and listening are acts of dynamical measurement. We continuously sample the incoming symbolic flow. We reconstruct a trajectory through what may be called semantic phase space. By this I mean the space of possible meanings, continuations, associations, constraints, and interpretive pathways made available by the symbols as they arrive.

As we read or listen, the trajectory may converge toward a stable basin of meaning. It may drift. It may hesitate. It may jump suddenly from one basin into another. A conversation often feels like coupled motion between minds. A train of thought feels like a path. A scientific argument feels like a route that must hold together across many steps. Sometimes, when the necessary basin is absent, an idea simply does not make sense. There is nowhere for it to land.

The previous essay gave us the sense of motion. It described language as a real-time, sequential, measurable flow. But it left unnamed the smaller working unit that makes those pathways possible.

Here I want to give that unit a name:

the finite functional symbolic trajectory, or more simply, the **Functional Symbolic Trajectory**.

What a Functional Symbolic Trajectory is

A functional symbolic trajectory is a finite symbolic pathway that carries representation, constraint, local uncertainty, and history through use.

It is not a static label or a perfect pointer to a separate realm of ideal meanings. Nor is it a fixed object sitting outside time. Rather, it is a stabilised series of symbols, actions, measurements, or formal moves that can be dynamically generated, followed, tested, remembered, compressed, extended, or allowed to fail.

Every word you read right now is part of a functional symbolic trajectory. Every sentence in an essay, every equation in a paper, every token generated by an AI model, every definition in a textbook, every measurement protocol in a laboratory, and every proof in mathematics can be treated in this way.

Each operates at a particular scale:

- A word is a small trajectory.
- A sentence is a larger trajectory.
- An argument is larger again.
- A theory, a discipline, or a culture of interpretation is larger still.

The important point is that these are not isolated objects. They are pathways that have to hold together in motion. A functional symbolic trajectory carries at least four things.

Representation is how the symbol fits into the wider semantic phase space. A word does not mean by itself. It means by the way it sits among possible uses, histories, contrasts, expectations, and continuations.

Constraint is the local rule structure that keeps the path admissible. This may be grammar, social agreement, mathematical formalism, experimental protocol, measurement convention, or the practical rules of a particular language game.

Local uncertainty is the neighbourhood of possible reconstructions and continuations surrounding the current symbol. A word always carries nearby alternatives. It has a region of possible meanings, not a single perfect point.

History is the generative path that brought the symbol into being. Much of this history is usually compressed or lost once the symbol is instantiated. A reader sees the word, not the whole path by which the word became useful.

Once symbols are seen in this way, representation and meaning begin to look less like fixed correspondence and more like a finite geometry of motion. This leads to the next idea i.e. a new functional symbolic trajectory: the **Fractal Geodesic**.

The Fractal Geodesic

When we consider language as a nonlinear dynamical system, a fractal geodesic is a self-similar bundle of functional symbolic trajectories nested across multiple scales.

The word fractal is used because the pattern repeats at different scales. A word has internal history and local uncertainty. A sentence has the same. A paragraph has the same. A theory has the same. The structure repeats, but not as a perfect copy. It repeats as a family of related symbolic movements.

The word geodesic is used because a coherent symbolic pathway often behaves like a path of constrained movement through a space of possibilities. It is not necessarily the shortest path in a simple geometric sense. It is the path that holds together under the constraints of the system.

- A good sentence is a small geodesic.
- A good proof is a formal geodesic.
- A good scientific theory is a large geodesic.
- A conversation that remains coherent is a coupled geodesic between speakers.

On Coherence

We can now see that coherence arises when the nested sub-trajectories flow without destructive conflict. They do not pull the reader into incompatible basins or introduce hidden assumptions that break the path. Rather, they remain mutually reinforcing with the larger trajectory they compose. Inversely, incoherence appears when a sub-trajectory slips its constraint, pulls against the parent path, or collapses into a nearby attractor that distorts the trajectory and reduces the functionality. This is why a single word can destabilise a sentence and why one hidden assumption can weaken a proof. It is also why one undefined term can make a theory harder

to follow and why an AI response can begin coherently and then quietly drift into what we call ‘hallucination’.

Historical provenance

The idea presented here did not appear out of nowhere. Rather, it sits inside a much longer historical trajectory concerning meaning, logic, language, information, and dynamical systems.

Bertrand Russell, in his work on logical construction, tried to clarify how many mathematical and conceptual objects could be treated not as directly given things, but as constructions that allow reliable work to be done. His famous programme of replacing inferred entities with logical constructions was an attempt to make the scaffolding more explicit.

In plain terms, Russell was trying to show how some “objects” used in language and thought may function as useful symbolic constructions rather than as directly measured things.

This is close to the spirit of the functional symbolic trajectory, although not quite the same. Russell was still working within the logic and foundations available to him. He did not have access to the later concepts and vocabulary of compression, nonlinear embedding, computational language models, signal reconstruction, or finite measurement geometry. He could see that some symbolic objects do work, yet could not yet describe them as trajectories through a dynamical semantic space.

Ludwig Wittgenstein, who studied with Russell, later moved the discussion in another direction. In his later writings, he described language as a vast collection of practices, often called language games. Words do not have meaning only because they point to things. They have meaning because of how they are used within local forms of life, local rules, and local contexts.

- The same word can do different work in different games.
- The meaning of a word depends on the path it is travelling.

In the language being developed here, many of Wittgenstein’s observations can be reinterpreted as observations about functional symbolic trajectories. He saw that words act at different scales. He saw that meaning depends on use. He saw that context is not decoration added after

the fact. Context is part of the pathway that makes the word function. This is why a word can be stable in one setting and unstable in another as the constraint structure has changed and the trajectory has moved.

Connecting Information to Mathematics

Claude Shannon changed the landscape again in 1948 with his mathematical theory of communication. Shannon was not trying to solve meaning in the ordinary human sense as his work was aimed at communication, uncertainty, coding, transmission, and noise. However, he gave us an engineering language for thinking about information as something that can be encoded, compressed, transmitted, corrupted, and recovered. This matters here because symbolic trajectories are not merely philosophical. They are also practical as they are compressed, transmitted and reconstructed and they are also vulnerable to noise and loss.

When a sentence is written, the full living thought that produced it is compressed into marks. And when a reader reads the sentence, the marks are expanded again into a reconstructed pathway. The original trajectory is never perfectly recovered, rather it is measured and measurements have uncertainty: they are part of an interactive dynamical process of measurement.

The Rise of Nonlinear Dynamical System Theory

Edward Lorenz then showed, in the early 1960s, how deterministic systems could behave in ways that were highly sensitive to initial conditions. His work on weather helped establish what later became widely known as chaos theory. A system could follow rules and still become difficult to predict over time.

This is important for language because language is not just a list of symbols. It is a sequential system with memory, feedback, instability, and basin changes. A single phrase can redirect an argument. A small ambiguity can produce a large interpretive change. A hidden assumption can quietly move the reader into a different attractor.

Finally, Floris Takens and others developed the modern mathematical toolkit of phase space reconstruction. Takens' embedding theorem showed that the hidden dynamics of a system can, under certain conditions, be reconstructed from a measured time series. This is one of the key bridges.

- Language arrives as a time series. Speech arrives through time.
- Text is processed through sequence.
- AI models generate tokens sequentially.

Meaning and representation, therefore, can be approached as a reconstructed dynamical pathway rather than as a static lookup between words and definitions. From the trajectory being shown here we can see the chain of provenance of these ideas:

- Russell gives part of the logical provenance
- Wittgenstein gives part of the use-based provenance.
- Shannon gives the communication and compression provenance.
- Lorenz gives the nonlinear dynamical provenance.
- Takens gives the reconstruction and embedding provenance.

Now in the modern age of AI, large language models now make the whole issue visible in a new way, because they generate symbolic trajectories in front of us, one token at a time and this gives us new instruments to measure the trajectories of language .

The fractal scale in practice

Once we have the unit, the functional symbolic trajectory, we begin to see fractal geodesics everywhere.

Everyday language

Consider a simple sentence:

“The bank is closed.”

This is one visible trajectory inside the dynamical time series of reading or listening. Inside it sit smaller sub-trajectories. The word “bank” carries more than one basin. It may mean a

financial institution or it may mean the side of a river. The word “closed” also constrains the interpretation. The reader’s prior context adds further constraint. For example, if the previous sentence mentioned money, accounts, or opening hours, the financial basin becomes stronger. If the previous sentence mentioned fishing, flooding, or walking beside a river, the riverside basin becomes stronger.

The sentence feels coherent when the sub-trajectories converge without conflict and it becomes unstable when they pull in different directions. Now consider:

“The bank is sky.”

or:

“The bank is driving.”

Here the reader feels friction. The trajectory does not settle easily. The words can be parsed grammatically, but their functional pathways do not naturally reinforce one another. The reader searches for a metaphor, a joke, a mistake, a surreal frame, or some missing context. That search is not incidental; it is the dynamical reconstruction of meaning in action. We feel logic when the trajectory holds and confusion when the trajectory loses admissibility.

Mathematics

Importantly, mathematics does not stand outside language. Mathematics is supported by language, notation, axioms, rule systems, historical agreement, training, and shared procedures of use. It is a highly constrained language, but still symbolic and sequential.

A proof can be therefore be treated as an example of a fractal geodesic. Each definition, lemma, equation, substitution, transformation, and conclusion is a sub-trajectory operating under strict formal constraint. The proof remains coherent only while every sub-trajectory stays nested inside the parent rule structure.

- A single illegal move breaks the path.
- A hidden assumption changes the basin.
- A poorly defined term adds uncertainty.

- A symbol introduced without constraint becomes unstable.

In considering proofs as functional symbolic trajectories, this does not weaken mathematics, rather it helps explain mathematics' power. Mathematics works so well because it creates highly disciplined symbolic trajectories. It restricts the possible moves and compresses prior work into reusable forms. It then allows readers trained in the same system to reconstruct the pathway with unusually low uncertainty. However, even in mathematics, the symbols are not magic. They still require admissible use.

A proof is not merely a collection of marks, it is a path that can be followed and a functional symbolic trajectory.

Large language models

For those that have followed my work, large language models make functional symbolic trajectories unusually visible. An AI-generated response unfolds one token at a time and each token is selected from a local region of possible continuations. The model does not produce a complete meaning all at once. Rather, it generates a pathway through a high-dimensional space of learned relations. From this perspective we can see that each next-token choice can be seen as a micro-trajectory.

The whole response is coherent when those micro-trajectories remain nested inside the constraint basin established by the prompt, the prior context, the training distribution, and the local structure of the response. 'Hallucination' is what it looks like when one or more sub-trajectories drift. For example:

- A name may be generated that fits the local linguistic shape but not the external record.
- A citation may sound plausible but fail as a measurement.
- An argument may remain fluent while its constraint structure has already slipped.

The trajectory of language reconstruction continues, but the geodesic has crossed into a conflicting basin. This is why AI hallucination is perhaps seen not only as an error of fact but also a trajectory error where the symbolic path remains locally smooth while losing contact with the intended constraint.

It seems to me this also explains why when interacting with both an AI or a person the language we use needs great care. Each sentence does not simply ask for a response, it shapes the basin of possible continuation and constrains the trajectory before it begins.

Physics and metrology

Continuing the functional symbolic trajectory presented here; physics and metrology provide another example. The modern definition of the SI second, from the International System of Units, is a textbook case of a metrological fractal geodesic.

It begins with a physical process, the caesium-133 hyperfine transition. It then moves through a counted relation, 9 192 631 770 periods. That relation becomes a formal symbolic definition. The definition becomes realised in instruments and the instruments then support clocks. Finally, the clocks support laboratories, navigation systems, software, engineering, publications, and downstream physical models.

Each layer is a functional symbolic trajectory operating at a different scale.

- The physical transition is one trajectory.
- The counting convention is another.
- The formal definition is another.
- The laboratory realisation is another.
- The software implementation is another.
- The textbook explanation is another.

The entire construct remains coherent only while the nested trajectories remain anchored to the originating measurement process and to the agreed metrological procedure. Importantly, this does not make the SI second unreal. Quite the opposite as it shows how much work is required to make it stable. We can see that a unit of measurement is not merely a word. It is a maintained symbolic trajectory held in place by instruments, institutions, definitions, histories, and repeated use.

The same applies more broadly in physics. A published result is not just a number. Rather, it is a pathway from apparatus, calibration, measurement, modelling, statistical treatment, notation, interpretation, and social acceptance. When that pathway is clear, the result becomes easier to

evaluate and when parts of the pathway are hidden, compressed, or smuggled in as assumptions, the functional symbolic trajectory starts to become unstable.

Does any of this matter?

For me, the answer is yes. The tools that make these trajectories visible are relatively new. Compression algorithms, signal theory, nonlinear embedding, large language models, precise metrological anchoring, and modern computation have changed what we can see. Older philosophy often treated symbols as static pointers, names, propositions, or logical objects. That work remains important, and started the trajectory presented here, however, we are no longer limited to that framing and can ask a more practical question. Not only, “What does this symbol mean?” but also: “How is the symbol situated on a finite symbolic pathway?”

Within the context of both science and AI this shift matters as it may deliver diagnostic tools. For example, when reading a scientific paper, we can ask where the main functional symbolic trajectories are and which definitions are doing the heavy work. We can ask whether the mathematical pathway stays inside its stated constraints and where an interpretation is added. We can ask whether a term is stable or whether it carries unresolved local uncertainty.

Then when debugging an AI output, we can ask where the response first drifted and which sub-trajectory may have crossed into a neighbouring basin. Finally, we can ask whether the output remained fluent after it had already lost constraint.

It also seems to me that it then follows, that when developing a new theoretical term, we can ask whether the term has enough history, constraint, and use to hold meaning. This may help to question whether it is a premature noun, a slow noun, or a stabilising pathway that may become useful through repeated measurement and refinement.

Importantly, the term “**Functional Symbolic Trajectory**” is itself a slow noun. My hope is that its meaning will sharpen through use. It is offered for others to test and stabilise, and as a working expression for something that appears across language, mathematics, AI, metrology, and scientific modelling. It names the finite symbolic pathway that allows meaning to be

constructed and for me helps explain why some arguments hold together and others drift. It also may help explain why words can be powerful, unstable, compressed, inherited, and dangerous. It also helps explain why a symbol is not merely a mark, but needs a path.

A small exercise

If you have followed the path of these words and reconstructed the meaning, within your own landscape of language and image of nested trajectories connects, perhaps try it on the next thing you read and consider the following questions.

- What are the visible functional symbolic trajectories here?
- Which words or symbols are carrying the most weight?
- Where do the sub-trajectories reinforce the larger path?
- Where do they begin to pull against it?
- Where is uncertainty being honestly carried?
- Where has it been hidden inside a compressed term?

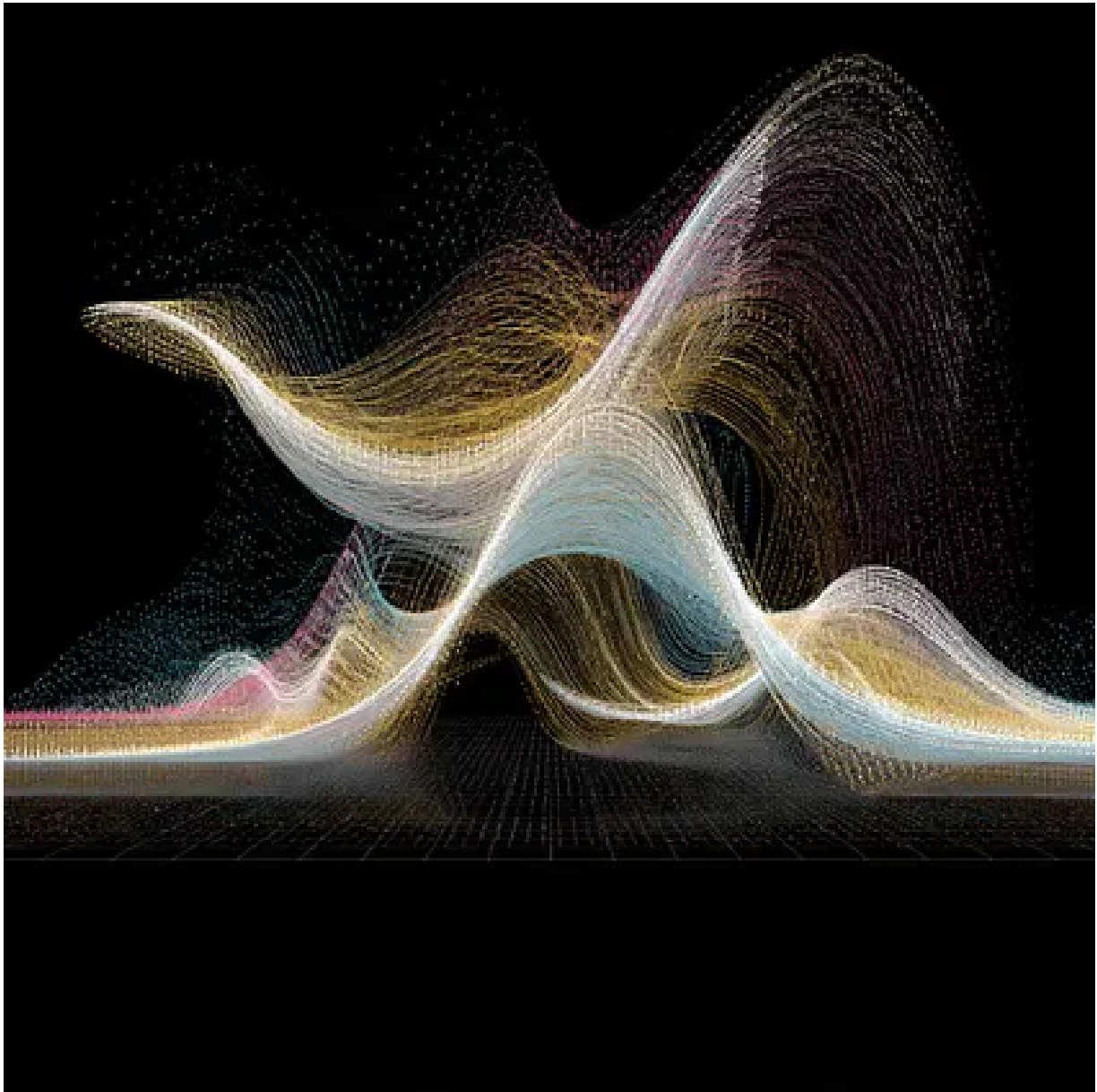
The exercise may be surprisingly revealing. A sentence stops being a row of words and becomes dynamic, as you observe the transition between words. You may begin to see that a proof stops being a static object and becomes a constrained path and an AI response stops being merely fluent or wrong and becomes a trajectory that can be examined.

As you read in this way, a scientific definition stops being a fixed statement and becomes a maintained symbolic instrument. And here we find the point of the concept, it gives us a way to see the motion - that is to say the dynamics of a living language that can never be trapped in stone.

Thank you for reading.

If you would like to follow the finer fractal trajectories, the related work below adds more technical depth. It remains a work in progress as I continue to explore which trajectories can be built, measured, stabilised, and tested within language, mathematics, AI, and scientific modelling.

Kevin



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Further Reading

[Full Essay on Functional Symbolic Trajectories](#)

[Finite Tractus: The Hidden Geometry of Language and Thought](#)

[Pairwise Phase Space Embedding in Transformer Architecture](#)

[Introducing the Takens-Based Transformer](#)

[JPEG Compression of LLM Input Embeddings](#)

[Introducing the Attralucian Essays](#)

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[Earlier works from a career in Biomedical Engineering](#)

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Omne quod est, finitum est; tantum per mensuram cognosci potest

Everything that exists is finite; it can only be known by measure

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